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Needfinding

Human Computer Interaction

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Hall of Fame or Shame?

A mix: it depends on the type of target using it



https://www.youtube.com/watch?v=e2RoNSKtVAo

Google Nest Mini: Issues

"Tap" is not for activating it!!

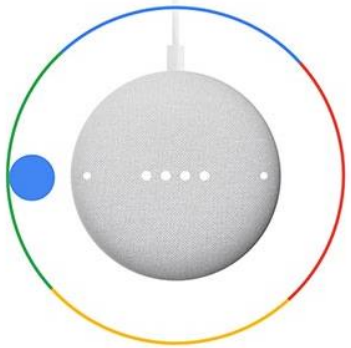
Play & pause

Tap the center of your Nest Mini



Change volume

Tap the right or left side to change volume. When something is playing, the lights will turn on as your hand gets close.



Mic switch

To turn off the mic, slide to show orange. The lights on top will also turn orange.



Others issues??

Possible solutions to the ”problems”

- An easier word to activate it
- More tangibility – activate it by touch
- Better onboarding – to learn how to use it
- Possibility to “learn” –es. singing a song for recognition



You need to understand the target of the product and the needs of the target

Goals

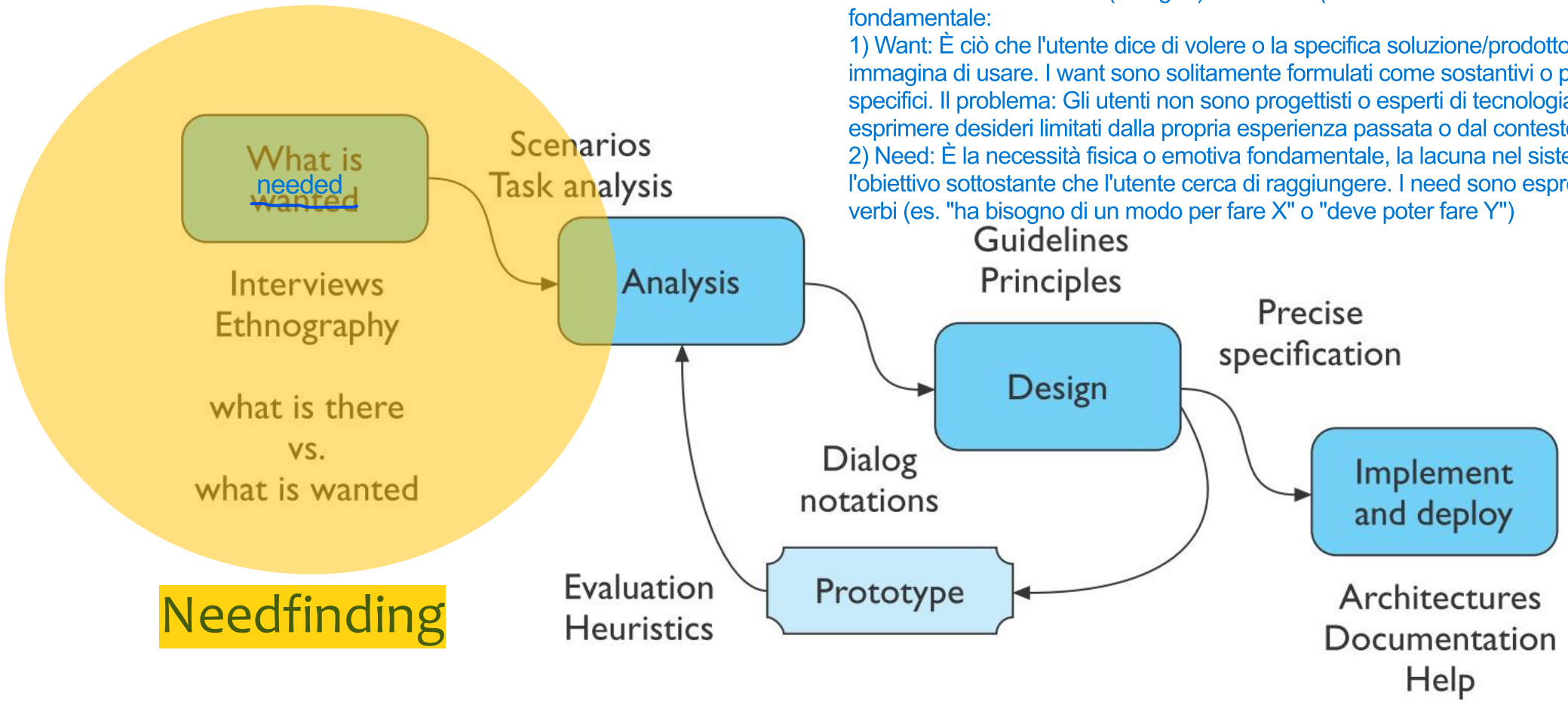
- ~~Finding and framing~~ Individuare e definire i **problems, starting from people**
- Understanding the system requirements and user needs
  - "Needfinding" Needfinding = Problem framing + understand what people need
- Knowing tools and ~~instruments~~ metodi for Needfinding

- Problem solving: you have a problem and you find a solution to the problem
- Problem framing: understand the problem (not finding the problem: Problem finding)

Human-Centered Design Process

Il compito di chi fa Needfinding non è chiedere agli utenti quale soluzione vogliono (perché spesso risponderanno basandosi solo su ciò che conoscono), ma osservarli e interrogarli per estrapolare il vero bisogno sottostante (il Need).

La differenza tra un Need (Bisogno) e un Want (Desiderio o Soluzione richiesta) è fondamentale:
1) Want: È ciò che l'utente dice di volere o la specifica soluzione/prodotto che immagina di usare. I want sono solitamente formulati come sostantivi o prodotti specifici. Il problema: Gli utenti non sono progettisti o esperti di tecnologia; tendono a esprimere desideri limitati dalla propria esperienza passata o dal contesto attuale.
2) Need: È la necessità fisica o emotiva fondamentale, la lacuna nel sistema o l'obiettivo sottostante che l'utente cerca di raggiungere. I need sono espressi tramite verbi (es. "ha bisogno di un modo per fare X" o "deve poter fare Y")



What Are **Needs**?

You need to write it down for the final project: you will find different needs from different people (usually, is not one person need, but you need to find the common needs between the people interviewed)

Definizione di Need

- "Human emotional or physical necessities"

Significa che un Need è uno stato di necessità della persona, che può essere:
- Fisico/pratico: la necessità di ridurre lo sforzo, il tempo, la fatica o l'incomodità (es. muoversi più velocemente, leggere senza sforzare la vista).
- Emotivo/psicologico: la necessità di sentirsi sicuri, in controllo, appagati o non frustrati durante l'uso di un sistema.
Mentre il gap è il problema nell'ambiente/sistema, questa definizione indica il disagio o la necessità reale provata dall'essere umano.

Gaps (gulfs) of Norman model

- "Gaps in a system"

C'è un divario (gap) tra ciò che l'utente vuole/deve fare e ciò che il sistema attuale (o la situazione attuale) gli permette di fare.

- They are **verbs**: activities and desires with which your user could use help
  - not nouns (solutions)

Need something (solution that there isn't) to do this (that is the need)

- Often, it is helpful to use the phrases 'needs a way to' or 'needs to be able to' to describe user needs
- They emerge directly from user ~~traits~~ or from contradictions between two ~~traits~~

caratteristiche

caratteristiche

  - such as a disconnect between what they says and what they do

"user traits" indica l'insieme di qualità fisiche, cognitive, demografiche e comportamentali che definiscono chi è l'utente.

Quando fai Needfinding, devi spogliarti dalle tue idee preconcepite, dalle tue ipotesi su quale sia il problema e da come vorresti risolverlo. Se entri in fase di ricerca avendo già in mente una soluzione (o un'idea di bisogno molto specifica), finirai per guidare l'utente verso la risposta che desideri tu.

A story…



wanted = solution; need = move faster

So…

- Is a faster horse a need???

So…

- Is a faster horse a need???
- NO! (there is no verb..!)
- It is a solution!! (and it is also a “want”)
- So.. Which is the **need**??

So…

- Is a faster horse a need???
- NO! (there is no verb..!)
- It is a solution!!
- So.. Which is the need??
  - Moving faster! (verb..!)
- So.. Why horses?
  - Because people know that!! Their experience is on that!
  - Note that cars are not the only solutions..! Different ways to solve the same needs
  - And this will happen to you as well!! Since they are going to talk about what they know! What they experienced! They are going to tell you about "their horses"

You need to find the needs of the target, then you will find the solution with your own problem solving

What is Needfinding?

Sono i gaps (gulfs) del Norman model

- Needs: **gaps** in a system

- Needfinding: discovering opportunities by **recognizing** this gaps



What is Needfinding?

So, Needfinding and then create the solution to solve the need

- Figure out the story of **what** and **why**…
- … and tell a new one!



Very important!!

- Needfinding means finding the needs, not the solutions!!

Prima di iniziare il Needfinding ti servono:
- un dominio di interesse: definire la macro-area su cui vuoi indagare (es. "la fruizione dei trasporti pubblici" o "lo studio in biblioteca").
- un'ipotesi iniziale per il reclutamento: una guida generale per sapere quali persone iniziare a contattare o quali ambienti osservare per primi.
Cosa cambia prima e dopo il Needfinding:
- PRIMA: Le risposte a Who, How e Context sono semplici ipotesi lavorative necessarie solo per perimetrare la ricerca.
- DURANTE e DOPO: Quelle domande diventano il tuo strumento di indagine per raccogliere dati reali, validati ed emergenti dal campo.

Iniziare credendo di avere già le risposte significa cadere nella trappola delle assumptions e rischiare di progettare per dei preconcetti anziché per i reali bisogni dell'utente. La fase di Needfinding (tramite osservazioni e interview) serve proprio a verificare, correggere o ribaltare queste supposizioni di partenza trovando la realtà dei fatti.

Main Needfinding Questions

- Needfinding = **Finding Potential User Needs**
  - What do users need?

Il Needfinding non serve a inventare prodotti, ma a scoprire opportunità nascoste identificando ciò di cui gli utenti hanno davvero bisogno prima ancora di progettare.

farsi queste domande prima è indispensabile per inquadrare il progetto, definire il team e scegliere la direzione. L'importante nel Needfinding è considerare le risposte iniziali come delle ipotesi di partenza che vanno poi confermate, smentite o arricchite dall'osservazione diretta degli utenti reali.

- That also requires
  - Who are the users? (age, capabilities, background, experts, etc.)
  - How are they doing it, now? (es. Commuting.. How? Buses? Trains? Single Ticket? Subscription? etc.)
  - What is the context in which they are doing it? (es. tourists, eramus students, etc.)
  - Can't we just ask them?
    - Hypothetically yes.., but.. the results could be confusing.. Too vague --- We need a more structured process!! -> needfindng!

Devi definire le loro caratteristiche specifiche

(the target depends on the macro idea that you have as project)

in the domain

the smaller it is the target, the easier will be to understand their need

Devi studiare il loro comportamento e le loro abitudini attuali.

Esempio della slide (Pendolarismo): Come viaggiano oggi? Prendono il bus o il treno? Usano biglietti singoli o abbonamenti? Usano app o biglietti cartacei?

L'ambiente e la situazione cambiano radicalmente il bisogno.

Un turista straniero che deve prendere il bus a Bologna ha bisogni diversissimi da uno studente Erasmus o da un residente che fa quel tragitto da 5 anni.

We will ask but before we need an idea as project team and then ask them structurely

Gli utenti non sanno esprimere i propri bisogni in modo razionale e tendono a chiedere solo "soluzioni" che già conoscono (devi aiutarli a trovare una soluzione)

La maggioranza di utenti che intervisti sono gli "standard users" per capire il comportamento tipico, ma intervisti anche utenti "imperfetti", non-utenti o casi limite perché sono loro a farti scoprire i bisogni più nascosti e le criticità più profonde del sistema.

Know Your Users (1)

«Know Your Users»



Inv c di far c ntinaia di int rvist sup rficiali a p rson a caso, il N dfinding punta a condurr indagini in profondità con un num ro limitato ma s l zionato di ut nti rappr s ntativi (r pr s ntativ us rs).

- Who are the users of the system?
  - Uniform, or different categories/groups?
  - Young/old? Novice/experienced?
  - Do not think of "generic" users, split the categories

suddividere l'insieme degli utenti in sottogruppi specifici e distinti, in base alle loro caratteristiche.

- *You* are not a [representative] user

If you find a problem in a domain, you need to confirm it with people (you don't have to solve your problem but user problem)

  - Designers and developers' skills, knowledge, attitude, background, interests, … are totally unlike those of your users
  - Except by chance (e.g., you are also students, developers, …)

si intende chi ha commissionato il lavoro

- The **client** is not a [representative] user
  - Bosses, managers, directors, … believe they know their employees and their jobs. Actually, they don't
  - Always seek the actual users that will use the system

It is costly and takes time

Know Your Users (2)

- Talking to users
  - Surveys (send around to get more and specific data, less direct)
  - Interviews costly and take a lot of time (we will focus on this)
  - Direct involvement (participatory design) we will not doing it: takes a lot of time (user inside the process). Generally, It is done with a very specific target
  - Bypass corporate policies
  - Understand real current behavior, pain points, workarounds, …
- Watching users we will not doing it
  - Observation sessions
  - Video recording (and analysis) privacy issues
  - Diaries analyse the notes taked
  - Analyze their work (artifact, processes, action sequences)
  - Discuss with users the findings of the observation (may discover the "why")

Co-design (workshop of 2 hours with target): similar to participatory design (users design the product and you get the artifacts created)

Know Your Users (3)

- Imagining users — You need to be an expert to use these methods
  - When real users are not available
  - Imagine how a real user would behave (very difficult) — Imagine the behaviour of users (target)
  - Building "imaginary" users: ***personas*** → 5/6 personas (standard users + users with special characteristics)
    - Detailed description of hypothetical persons in a given role
    - Imagine them as they were a real person

You can combine the methods (e.g. personas + interviews)

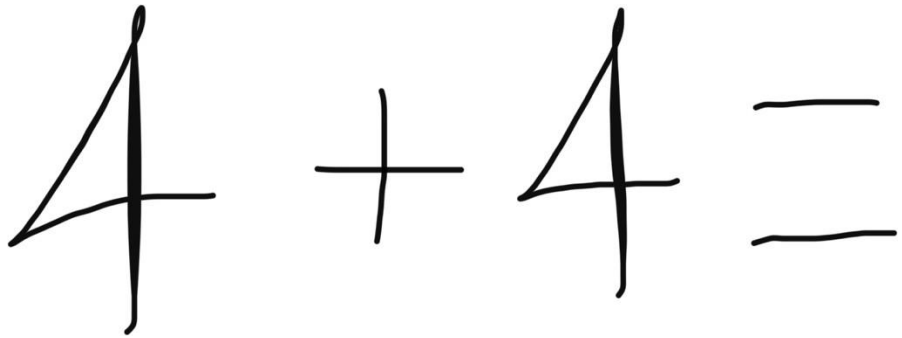
Problem Framing

Perché se ti limiti a trovare un problema (Problem Finding) ma non lo inquadri correttamente (Problem Framing), cioè rimani in superfici e non vai in profondità per capirlo veramente, rischi di progettare la soluzione perfetta per il problema sbagliato.

- Framing problems is as important as finding them
  - we need to check our and other people's **assumptions** to fully understand the needs, the context, and emphasize with them

Comprendi il contesto reale e i bisogni profondi: Cerchi di capire la situazione a 360 gradi prima di saltare alle conclusioni o proporre soluzioni.

- For example, let's say that a person shows this to us:

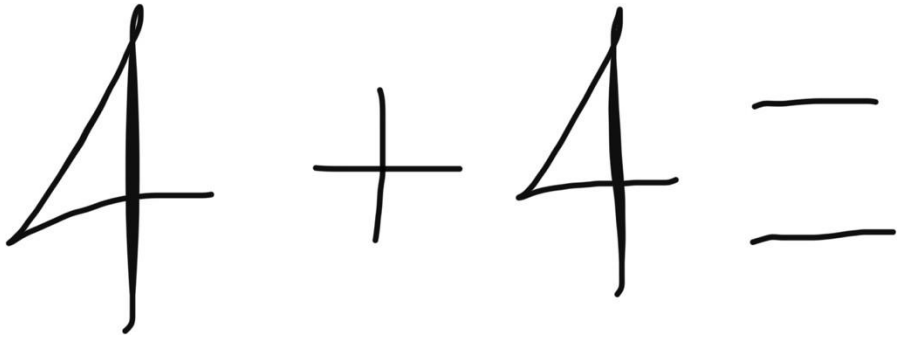


Verificare le supposizioni: Metti in dubbio ciò che tu, il tuo team o il cliente date per scontato. Chiediti sempre: "Stiamo dando per scontato qualcosa che in realtà non è vero per l'utente?" (come nell'esempio 4 + 4 =, dove diamo per scontato che siano numeri aritmetici).

Empatizzi con le persone: Ti metti nei panni dell'utente per vedere il mondo dal suo punto di vista e percepire i suoi reali stati d'animo.

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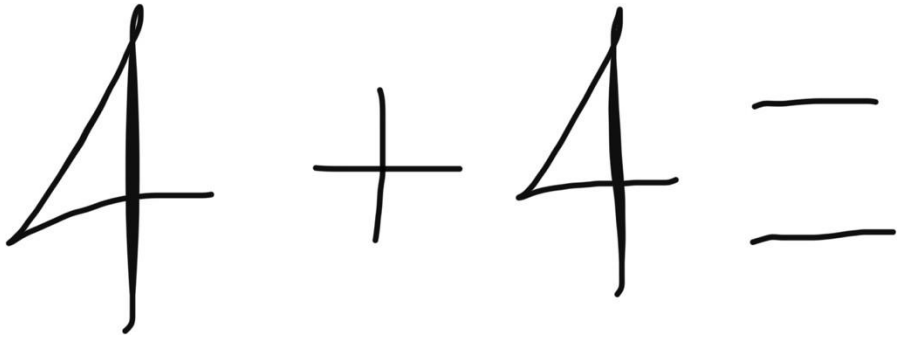
Senza Problem Framing: Vedi 4 + 4 = e la tua mente assume subito che si tratti di un'addizione aritmetica, per cui concludi che il problema sia trovare la somma (8).
Con il Problem Framing: Ti fermi e metti in discussione le supposizioni (check assumptions).

In sintesi, il Problem Framing è l'atteggiamento mentale di non dare mai nulla per scontato e analizzare i fatti senza i propri preconcetti.

Assumptions!!!
You assumed that those characters are numbers, and the plus is the arithmetical sum.. But.. You don't actually know!

Problem Framing

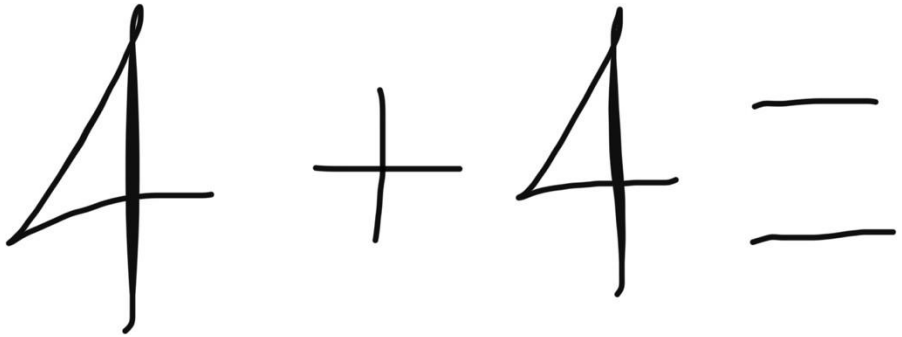
- Framing problems is as important as finding them
  - we need to check our and other people's **assumptions** to fully understand the needs, the context, and emphasize with them
- For example, let's say that a person shows this to us:



Questions should be:
- Are those numbers?
- Are those symbols in the arithmetical domain?

Problem Framing

- Framing problems is as important as finding them
  - we need to check our and other people's **assumptions** to fully understand the needs, the context, and emphasize with them
- For example, let's say that a person shows this to us:



You need to be very careful!! Assumptions can *break* your needfinding process!



Needfinding Methods

Techniques adopted in Needfinding and Requirement Analysis

Many Methods!

- Observations
- Diaries — we will not use them here (but it can be useful)
- Interviews
- Focus groups — Similar to co-design; we will not use them here (but it can be useful). Can be used at the beginning or during the process. Start discussion with people, show the prototype and get feedbacks
- Surveys
- Contextual Inquiry



Interview last between 30 and 60 minutes

Can be used for whatever needfinding method

To Get Started…

For us, is the interview

After having identified and found the users to <*insert needfinding method*>:

1. Introduce yourself and try to make the participant comfortable.

and yourself

2. Discuss confidentiality and get *explicit consent* for any filming or recording you may be doing.

   1. We need to be complian with EU *General Data Protection Regulation* (GDPR)
   2. *Written Or recorded*

Interviewed consent can be

   3. *For children, you need the signed consent from parents/legal guardian; the child still needs to give you the assent (interest in participating)*

(the users need to be able to quit the interview, if they are no more comfortable: must be said at the beginning)

3. Begin to introduce the topic/goal you are interested in understanding.

(Why are you there?)

Understand the need in a specific domain

4. Start <*insert here needfinding method*>.

For us, is the interview

Observation

Extract information from people without explicitly asking (by observing people)

"You can observe a lot, just by watching" – Yogi Berra



The most known method of Observation

We are not doing it (it is difficult)

Ethnographic Observation

L'osservazione etnografica è un metodo di ricerca qualitativa delle scienze sociali che consiste nello studiare un gruppo, una comunità o un comportamento umano inserendosi direttamente nel suo ambiente naturale

Immergersi

- ~~Embed~~ in the users' environment, culture, behavior — Done to try to understand some kind of interactions between people and between people and environment, etc etc
- Goal: to obtain the necessary data to influence interface (re)design
- Learn the language of users and their environment
- Listen and observe carefully — Example: observe and take notes about people interaction with vending machine
  - A volte
  - ~~Sometimes~~ ask questions and clarifications — if there is a person acting differently (generally, you don't approach people)
- Audio-Video Record / Take Notes — When it comes to recordings, you must ask for permission to record for privacy reasons, but there's no need to ask when taking notes
- Risks:
  - Misinterpret observations
  - Disturbare o alterare la normale pratica quotidiana
  - ~~Disrupting normal practice~~ — si riferisce al fatto che la sola presenza dell'osservatore nell'ambiente dell'utente può alterare, interrompere o modificare il modo in cui l'utente svolge normalmente le sue attività quotidiane.
  - Trascurare/Ignorare
  - ~~Overlooking~~ important information — (può succedere anche il contrario: cioè troppo focus su dettagli considerati importanti dell'osservatore, ma in realtà non lo sono)

What Should Learn By Observation? (1)

Queste cose si osservano e si analizzano SEMPRE all'interno di uno scenario e di un contesto specifico, mai in generale! Quindi, queste domande vanno fatte/pensate in un certo scenario/contesto

1. What do people do ~~now~~? (attualmente)
2. What values and goals do people have?
3. How are these activities ~~embedded in a larger ecology?~~ (inserite in un'ecologia più ampia?) (ovvero in un ecosistema/sistema più vasto di relazioni e strumenti)
4. Similarities and differences across people
5. Other types of context, like time of day — The situation change based on context (the interaction changes?)?

Especially tacit (unspoken) knowledge — la conoscenza non espressa a parole

What Should Learn By Observation? (2)

Processo vs. Pratica

- Process vs. Practice

You observe Processes and Practices of people without asking (si va a distinguere ciò che dovrebbe accadere in teoria da ciò che accade realmente nella pratica quotidiana)

- Process: how things are officially supposed to happen, and are officially part of the training
- Practice: set of workarounds, practical tricks, information learnt from the field and from experience, etc., that are part of daily activities

Types of Observation (1)

How to Conduct User Observations
https://www.interaction-design.org/literature/article/how-to-conduct-user-observations

Example of Hospital: to observe a specific situation, we reproduce in our company the Hospital environment (and you analyse how they interact with the system). It is costly but it is necessary

Di solito, siamo in una grande azienda e c'è uno spazio specifico per fare osservazione dove le persone sanno di essere osservate

Controlled Observation, within a Lab environment

- **Easy to reproduce.** If you use a quantitative approach – easy to get similar results by repeating
- **Easy to analyze.** Quantitative data requires less effort to analyze than qualitative data.
- **Quick to conduct.** Recruitment may take a little time, the controlled observation is fairly fast to run.
- **The Hawthorne Effect.** The act of observation of how someone does something can change their approach to carrying out the task.

ideazione: processo mentale con cui si concepiscono, formano e sviluppano nuove idee o progetti

Types of Observation (2)

Naturalistic Observation: studying the user "in the wild", less structured

- **More reliable**. When people use a product in real life – they are much more likely to encounter the frustrations (and benefits) of real life use than they are in a lab following a set of instructions.

affidabile

- **More useful for ideation**. Qualitative research can generate lots of ideas for product improvement as it opens up possibilities that aren't found in quantitative research.
- **Difficult to include a representative sample.** More expensive and time consuming to conduct, limits the reach of the research. Use this kind of research to create ideas and then test those ideas with other forms of research.
- **Difficult to make them replicable.** Problem with sample sizes and dependence on the observer.
- **Hard to manipulate external variables.** For example, if it's raining when you observe your users working on a smartphone – their behavior is likely to be different to when it's sunny. You have no control over the weather "in the wild".

The context change (e.g. sunny and rainy day)

Come osservare le persone senza alterare il loro comportamento (evitando l'Effetto Hawthorne).

Mimetizzarsi

~~Blending In~~

ruolo completamente passivo

- **Becoming part of the wall**

(Complete Observer) No one notice you

- Avoid being intrusive or modifying behaviors
  - Avoid video-recording or interruptions
- Programmare del tempo
- ~~Schedule time~~ for discussing your observations

Obiettivo: Far dimenticare all'utente la tua presenza in modo che agisca in totale naturalezza.

il ricercatore si immerge attivamente nel contesto, quasi come un "agente sotto copertura" o una spia

- **Becoming "one of them"**, like a **spy**

(Complete Participant) You become part of the group to understand better how they act

- Sottoporsi al
- ~~Undergo~~ training process
  - Official information
  - Matter-of-fact information shared by co-workers
- Observe all the practices
- Validate your observations with the others

Obiettivo: Comprendere dall'interno la cultura, il linguaggio e la realtà operativa che dall'esterno non sarebbero mai visibili.

In generale, non solo Observation

Data Collection

Questa slide tratta della classificazione dei dati che un ricercatore raccoglie durante la fase di Needfinding.

Subjective

- Impressions
- Ranking/ratings by users on different questions (If you ask questions to users)
- Written summary report (report scritto dal ricercatore)
- Artifacts and "hints" ~~in the~~ sul ~~workplace~~ posto di lavoro (se, ad esempio, si coinvolge l'utente nel Co-design, vengono prodotti degli artefatti)

Objective

- ~~Anecdotes~~ Aneddoti/Episodi concreti
- Critical incidents
- Observed errors
- Observed ~~workarounds~~ scorciatoie



Interviewing

Asking users about their needs and desires…

…what could possibly go wrong?

Main Forms of Interviews

- In-person interviews
  - Time-demanding, in-depth knowledge
  - Structured vs Unstructured
  - One-to-one vs. Focus groups
- Surveys (you need a lot of data to be reliable)
  - Fast, more superficial
  - Sets of questions with predefined possible answers
  - Paper-based or on-line (less time consuming than interviews)

consente di scavare a fondo nella mente dell'utente.

- Structured: list of questions to ask (same questions to all users)
- Unstructured: (more complex) start talking and questions asked driven by users' answers (so different questions for users)
- Semi-structured: mix of two methods (some question common for all users and some specific for each user based on their answers)

Interview in group (problema: alcune persone (con personalità forti) parleranno molto, mentre altre (timide) parleranno meno)

Gli utenti provano frustrazione o disagio quando usano un sistema, ma non riescono a tradurre questa sensazione in una richiesta razionale o in una soluzione. Percepiscono il problema (Need), ma non sanno come esprimerlo a parole.

Beware!

- Users don't know what they want

  (they think they know what they want; they know the issue, the problem, the gap; They probably know the need, but they will try to tell you what they want (WANT is not what we want to understand: we focus on the NEEDs))

  - Maybe subconsciously, but not rationally
  - They will tell what they think you like to hear — They think that you want to hear what they want, not their needs
  - Especially for "new" products or "disruptive" technologies — Asking questions about it, is not so easy
  - They lack the creativity or the technical expertise to understand the new product — Gli utenti sono esperti del loro lavoro/dominio, non sono designer né sviluppatori di software. Non conoscono le potenzialità della tecnologia attuale né i principi di interfaccia. Chiedere a loro come progettare un sistema è come chiedere a un paziente come il chirurgo dovrebbe operarlo.
  - They take the current context for granted (e.g., required workarounds "because, yes") — Le persone si abituano alle inefficienze fino a non notarle più. Se una procedura è lenta o assurda, dopo mesi o anni che la ripetono diventa per loro "normale". Usano scorciatoie e trucchi creati da loro senza nemmeno rendersene conto, rispondendo "Si è sempre fatto così" o "Perché sì". Poiché hanno normalizzato il problema, non te lo segnaleranno mai come un bisogno durante un'intervista.

Le persone riescono a immaginare solo piccoli miglioramenti di ciò che già conoscono. Non riescono a immaginare qualcosa che non esiste ancora. Il famoso esempio di Henry Ford: "Se avessi chiesto alle persone cosa volevano, mi avrebbero risposto: 'Cavalli più veloci'". Nessuno avrebbe chiesto un'automobile, perché non la conosceva. Prima dell'uscita dell'iPhone, se avessi chiesto agli utenti cosa volevano cellulari con schermo touch, ti avrebbero chiesto telefoni con la tastiera fisica leggermente più grande.

Stop Asking Users What They Want
https://uxplanet.org/stop-asking-users-what-they-want-21e9ba646bce

Come reclutare e scegliere le persone giuste da intervistare. Se scegli i partecipanti sbagliati, rischi di raccogliere informazioni fuorvianti che comprometteranno l'intero progetto.

Choosing Participants For Interviews

- Representative of target users
  - [All] Stakeholders

- Rappresentativi del target: non devi intervistare persone a caso o "utenti generici", ma persone che possiedono le reali caratteristiche, competenze e vincoli della popolazione a cui ci si rivolge.
- Tutti gli Stakeholder: Gli utenti non sono solo chi preme fisicamente i tasti. Devi includere tutte le figure coinvolte direttamente o indirettamente dal sistema.
> Esempio: Per un software di gestione medica in ospedale, gli utenti diretti sono i medici e gli infermieri, ma gli stakeholder includono anche il personale amministrativo e i responsabili dell'IT. Ciascuno ha bisogni e punti di vista differenti.

Potrebbero essere

- ~~May be~~ current users of a similar system

Se stai riprogettando un software esistente o creando un prodotto concorrente

Potrebbero anche essere

- ~~Might also be~~ the non-users (for a new product)

Intervistare chi NON usa il prodotto attuale o rifiuta la tecnologia è preziosissimo. Ti aiuta a capire le barriere all'entrata: perché rifiutano il sistema? Quali paure, difficoltà o pregiudizi hanno? Quali metodi alternativi/analogici usano per farne a meno?

- Approximate, if necessary (with similar users)

Se l'utente sa fin dall'inizio che riceverà un regalo (o peggio, se lo riceve prima dell'intervista), scatta un meccanismo psicologico inconscio:
- Senso di debito/gratitudine: L'utente si sente in dovere di "ricambiare la gentilezza".
- Paura di deludere: Tenderà a fare complimenti al sistema, a dire che l'interfaccia è bella e facile, e nasconderà le sue reali frustrazioni o gli errori commessi.
- Risposte "giuste" anziché "vere": Cercherà di indovinare cosa il ricercatore vuole sentirsi dire.
In molti protocolli di ricerca, si dice all'utente che parteciperà a una breve sessione di studio e, solo al termine dell'intervista/osservazione, gli si consegna il piccolo regalo o la carta regalo come "ringraziamento per il tempo dedicato".

- Incentives, motivations, small gifts

Esempio: Se il tuo progetto è un'app per la mobilità sostenibile (es. car-sharing), si intervistano 4 persone:
- 2 Target Users: Persone che usano già regolarmente servizi di mobilità in sharing (ti dicono cosa funziona e cosa no nell'esperienza attuale).
- 2 Non-Users (ma "Closeby"): Persone che si spostano tutti i giorni in città ma usano solo l'auto di proprietà o il mezzo privato.
A cosa servono: Ti spiegano le barriere all'ingresso: perché non hanno mai scaricato l'app? Cosa li spaventa? Quali pregiudizi hanno?

Executing Interviews

- Schedule a time and place comfortable for users
- Introduce yourself, explain your purpose
  - You are not testing them; they are helping you (no wrong/correct answer to questions)
- ~~Begin with open-ended, unbiased, non-leading questions~~ Iniziare con domande a risposta aperta, neutre e non pilotate
- Ask the question and let them answer
  - Give enough time. The 2nd reply is often more interesting than the 1st.
- Follow-up with related questions. Deep dive into interesting points



Interview
http://dlrtoolkit.com/interview/

Guidelines For Questions

- Structured questions are easier to process (per chi fa l'intervista), unstructured questions ~~solicit~~ (sono più complesse da analizzare) ~~more comments~~ It is better to have semi-structured interview: a mix of structured and unstructured
- ~~Open-ended~~ questions (aperte), with follow-up discussion
- For quantitative questions (e.g., rate in a scale 1 to 5, such as Likert), always ask what they mean by "4" Why 4? So, you have a mix of number and description of the vote
- Aim at direct, concrete, specific questions that ask for detailed answers
- Use the (context) language of the user
- Always try the question with a smaller (trusted) group, for debugging (to understand if the question is good)

Examples of Open-Ended Questions

Structured questions

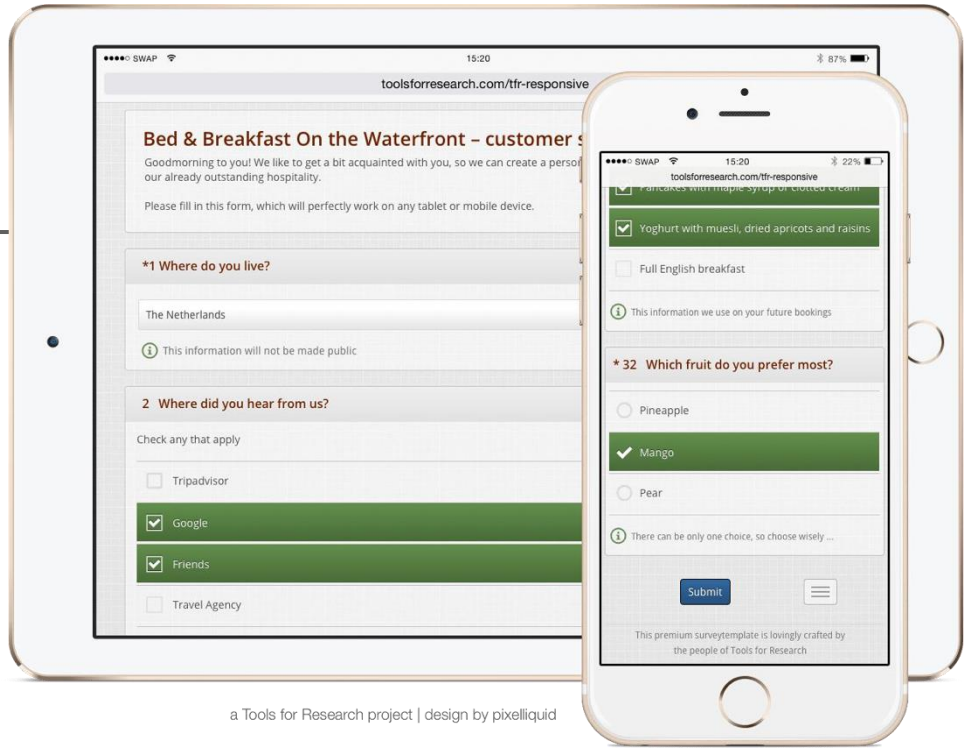
- ‘Tell me about your typical day.’
- Tell me three good things about. .. ’ [always start with positive things!!]
- ‘and three bad things.’
- ‘What has gone wrong with the application recently? How did you cope?’
- ‘What else should we have asked about?’ [to conclude..]

Bad questions – To Avoid

- Is feature [x] important to you?
  - 'Leading' question [it implies that the activity is important and also avoid yes/no questions]
- What would you like in a tool? Sarebbe come chiedere "What do you want?"
  - User are experts in their domain, not expert in design
- What do you like in [x]?
  - Assuming question. Maybe he/she doesn't like it [or he/she loves it]
- What would you do in a hypothetical situation?
  - Users cannot imagine the complete environment or an unusual situation

Bad questions – To Avoid

- How often do you do [x]?
  - Humans are very bad at estimating (and biased in the answers)
  - May obtain by log analysis (if an application already exists)
- Binary questions (yes/no)
  - Do not yield motivation
- 'Tell me a story about you'
- 'How do you reach the decision? Did someone decide without you? …'
  - Avoid suggesting possible answers. Trust the question: ask it and wait in silence



Surveys

Requirements gathering through (on-line) surveys and questionnaires

On-line surveys

- Familiar, cost-effective, potentially reaching a very wide audience
- Results can be easily visualized and analyzed with statistical methods

- Should be very careful with setting the goal of the survey
  - First define what statistics and/or charts you need, then design the survey structure and questions and then validate the survey to see if the questions are statistically meaningful

Risks in surveys

- Good for a ~~shallow view over a large base of respondent~~ (visione di superficie su un'ampia base di rispondenti), but not for a “deep” analysis
- Impossible to ask follow-up questions (domande di approfondimento)
- Biased data if questions rely ~~on user’s memory~~ (sulla memoria dell'utente) or ~~on~~ (su) ~~“sensitive” issues~~ (argomenti sensibili) (money, emotions, …)
- Finding ~~a representative target population~~ (una popolazione target rappresentativa)
  - Stratification

Not a big issue: there are companies that can find your target population for your survey

Survey structure

- Declare the purpose of the survey and the expected time
- One or more sections with targeted questions
- Background information about the users
  - Limit “mandatory” fields to a minimum



Important for analysis for a specific group (e.g. age, gender, etc etc)

Background information (examples)

Some data are sensitive (GDPR compliance)

- Background demographics (age, gender, origins, native language, education, income)
- Experience with computers (specific applications or software packages, length of time, depth of knowledge, whether knowledge was acquired through formal training or self-teaching)
- Job responsibilities (decision-making influence, managerial roles, motivation)
- Personality style (introvert versus extrovert, risk taking versus risk averse, early versus late adopter, systematic versus opportunistic)
- Reasons for not using an interface (inadequate services, too complex, too slow, afraid)
- Familiarity with features

Types of Questions

- Open-ended questions
  - ~~Solicit~~ Ask specific motivation, to avoid too generic answers
  - Require methodology for analyzing the responses
- Closed-ended questions
  - Only one possible choice
  - **Ordinal** values: a scale of ordered possibilities (e.g., from 1 to 5)
  - **Nominal** values: alternatives, with no ordering relationships (e.g., city or department)

Measurement scales (1)

- Nominal scale
  - Predefined set of distinct (pairwise disjoint) classes. No ordering.
  - Example: Gender. City of residence. Course of enrollment. Preferred color. Room of your house.
  - Meaningful statistics: Counting. Frequency. Mode.
- Ordinal scale
  - Distinct classes, pre-defined ordering. No defined "distance" across the different values
  - Example: University scores. Preference (**Likert**). Stars. Thumbs up/down.
  - Meaningful statistics: Ordering. Median. Quartiles. Percentiles. Rank.

scale 1 to 5

Examples of ordinal scales

How do you feel today?

- 1 – Very Unhappy
- 2 – Unhappy
- 3 – OK
- 4 – Happy
- 5 – Very Happy

How satisfied are you with our service?

- 1 – Very Unsatisfied
- 2 – Somewhat Unsatisfied
- 3 – Neutral
- 4 – Somewhat Satisfied
- 5 – Very Satisfied

Q. I take home-cooked meals to the office.

Answers:

Always	Often	Sometimes	Rarely	Never

Measurement scales (2)

- Interval scale (rarely used)
  - Numerical value, arbitrary zero-value, arbitrary unit of measure
  - Example: Date. Hour. Temperature.
  - Meaningful statistics: Mean (Average). Variance.
- Ratio scale
  - Numerical value, fixed zero-value, arbitrary unit of measure
  - Example: Duration of a task. Length of an object. Age.
  - Meaningful statistics: All (higher-order, geometric mean, …)
- Absolute scale
  - Numerical (integer) value, corresponding to the cardinality of a set
  - Example: Number of employees
  - Meaningful statistics: All

Examples of ratio scales

What is your weight in kilograms?

- Less than 50 kilograms
- 51- 70 kilograms
- 71- 90 kilograms
- 91-110 kilograms
- More than 110 kilograms

Interval scale	Temperature 0 degree 1 degree 2 degree . 100 degree	Grade or IQ score 0 score 50 score 60 . 100 score	Opinion/perception scale 0 1 2 . 100
Ratio scale	Height 0 feet 2.5 feet 5 feet 10 feet . 100 feet	Weight 0 Kg 5.5 Kg 10 feet 50 feet . 100 feet	Income 0 5$ 10.5$. 100$ 1000$

Ordinal scales: Likert scales

- Ask for the level of agreement about a statement
- Extreme values are very rare to be selected
- An even number of levels prevents a “neutral” response

- 4-level: Strongly Disagree / Disagree / Agree / Strongly agree
- 5-level: Strongly Disagree / Disagree / Neither agree nor disagree / Agree / Strongly agree
- Numerical ranges: 1-5, 1-7, 1-9



https://en.wikipedia.org/wiki/Likert_scale
https://www.simplypsychology.org/likert-scale.html

Deepening… Likert scale *

- **Can Likert Scales be Treated as Interval Scales?—A Simulation Study**

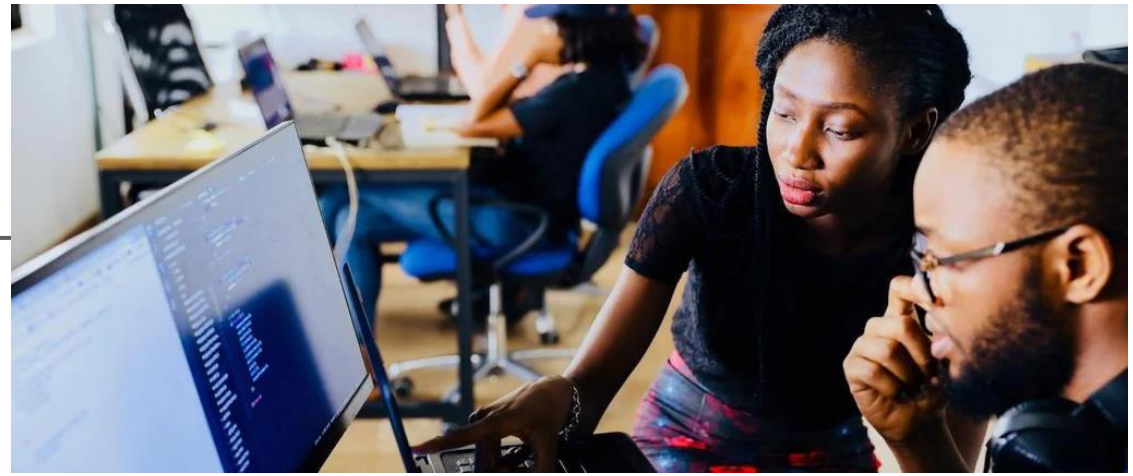
https://www.tandfonline.com/doi/full/10.1080/01488376.2017.1329775

Levels of Measurement

Offers:	Nominal	Ordinal	Interval	Ratio
The sequence of variables is established	–	Yes	Yes	Yes
Mode	Yes	Yes	Yes	Yes
Median	–	Yes	Yes	Yes
Mean	–	–	Yes	Yes
Difference between variables can be evaluated	–	–	Yes	Yes
Addition and Subtraction of variables	–	–	Yes	Yes
Multiplication and Division of variables	–	–	–	Yes
Absolute zero	–	–	–	Yes

Warnings

- If possible, use two simpler questions instead of one complex one (separate the factors of interest)
- Avoid negative words in the question
- Biased questions solicit biased responses



Contextual Inquiry

Observing while interviewing

Contextual Inquiry

- Contextual inquiry = inquiry of context
  - *Context*: the research takes place in the users' natural environment as they conduct their activities the way they normally would
  - *Inquiry*: the researcher watches the user as she performs their task and asks for information to understand how and why users do what they do
- Participants are observed while they perform tasks **and** talk about what they are doing while they perform them
  - Observation mixed with interviews
- Participants take a more *active* role in leading their session, like a craftsman with their apprentice



Contextual Inquiry Vs. Interviews

- Interviews and surveys rely on the users' ability to recall and explain a process that they are removed from in that moment
  - People attempt to summarize their processes, but important details like reasoning, motivation, and underlying mental models are *left out* of this summary
  - Often you only obtain a superficial understanding of the users' approach to the activity
- However, people can easily talk about **what** they are doing and **why**, when they are doing it
  - You get to see the interruptions, superstitious behaviors, and illogical processes

Contextual Inquiry Vs. Observations

- Contextual inquiry works well for understanding **in-depth thought processes** of users
  - and the underlying structure of their activities
- Direct observation works better when:
  - The activity does not require in-depth thoughts, e.g., designing a standard e-commerce page
  - People cannot (or should not) be interrupted or distracted
    - Think at doctors in hospitals, for instance
    - In these cases, a valid option is an observation followed by an interview

Risks and Downsides (I)

- **Participants always go into "interview mode"**
  - Working and demonstrating while being interviewed is unfamiliar to most people
  - It can be easy for participants to begin summarizing their processes at the end
- **It is a show-and-tell of frustrations with the current solution**
  - Participants may feel you are looking for their feedback on all the problems with the system
  - However, the purpose of contextual inquiry is to understand people's thoughts and work process, regardless of the solution

Risks and Downsides (II)

- **You introduce biases**
  - It is important to approach contextual inquiry objectively, without any preconceived notions or opinions about the subject matter
  - Go into the activity with an open mind, treating everything you learn with the same level of importance
- **You bias the user**
  - As you perform the contextual inquiry, it is possible that the participant may adjust their process to fit into the discussion or your interpretations

References

- David Benyon: Designing Interactive Systems, Chapter 7
- Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale: Human Computer Intera ction, 3rd Edition, Chapter 5
- Shneiderman, Plaisant, Cohen, Jacobs, Elmqvist & Diakopoulos: Designing the User Interface: Strategies for Effective Human- Computer Interaction, 6th Edition, Chapter 4, 5.4
- Lazar, Feng, Hochheiser: Research Methods in HCI, 2010, Chapter 9 (Observation), Chapter 5 (Surveys)
- Contextual Inquiry, https://www.nngroup.com/articles/contextual-inquiry/

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